

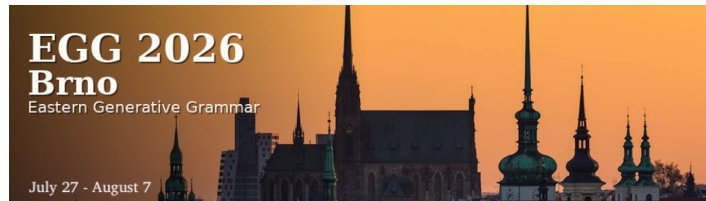


The Functional Architecture of Nominals

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<https://kxhoge.github.io/resources.html>



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Course outline

Monday	Approaches to the functional structure of nominals
Tuesday	The NP vs. DP debate
Wednesday	The NP vs. DP debate (cont.)
Thursday	Pronouns
Friday	D(emonstrative)-Pronouns

The NP vs. DP Debate

Outline

1. Determined and determinerless nominals in cross-linguistic perspective
2. An NP/DP parameter?
3. Evidence for DP-structure in articleless languages
4. Summary and outlook

The NP vs. DP Debate

Determined and determinerless nominals in cross-linguistic perspective

- Many languages lack definite and/or indefinite articles.

- (1) a. Kniha je na stole.
book is on table
‘The book is on the table.’
b. Na stole je kniha. [Czech]
on table is book
‘There is a book on the table.’
- (2) a. Orhan ev-i almak istiyor.
Orhan house-ACC buy want.PROG.3SG
‘Orhan wants to buy the house.’
b. Orhan bir ev almak istiyor. [Turkish]
Orhan a house buy want.PROG.3.SG
‘Orhan wants to buy a house’
(cf. Köylü 2021: 7, ex. (8))
- (3) a. Ég hef lesið bókina.
I have read book-the
‘I have read the book.’
(Thráinsson 2007: 6, ex. (1.13a))
b. Ég hef lesið bók. [Icelandic]
I have read book
‘I have read a book.’

The NP vs. DP Debate

Determined and determinerless nominals in cross-linguistic perspective

- The cross-linguistic data raise questions about the universality of DP-structure and the consequences of the presence or absence of the functional category D.
 - Does the absence of lexical items associated with D correspond to the absence of the syntactic category D?
 - Do determined and determinerless nominals differ in their syntactic and semantic behaviour?

The NP vs. DP Debate

Determined and determinerless nominals in cross-linguistic perspective

- Even in languages with definite and indefinite articles, we find bare nominals.

In English, bare plurals and mass nouns admit both generic (kind-oriented) and existential readings.

- (4) a. Vaquitas are endangered.
b. Vaquitas are frolicking in the water.
- (5) a. Gold is rare.
b. Gold was found in a Scottish river.



Definite plurals and mass nouns cannot freely receive kind-denoting interpretations.

- (6) a. *The vaquitas are endangered.
b. *The gold is rare.

The NP vs. DP Debate

Determined and determinerless nominals in cross-linguistic perspective

- Other languages resort to different strategies when expressing existential and generic readings of plural and mass terms.

In Italian, definite articles are used for generic readings, while partitive constructions are used for existential readings.

(7) a. I cani sono rari.
the dogs are rare
'Dogs are rare.'
(Chierchia 1998: 342, ex. (4b))

(8) a. L'oro è raro.
the-gold is rare
'Gold is rare.'

b. Dei cani stanno giocando fuori.
of-the dogs are playing outside
'Dogs are playing outside.'
(Chierchia 1998: 342, ex. (5a))

b. È stato trovato dell'oro in Scozia.
is been found of-the gold in Scotland
'Gold was found in Scotland.'

The NP vs. DP Debate

Determined and determinerless nominals in cross-linguistic perspective

- Chierchia (1998) interprets the contrast between English and Italian as evidence that languages differ in the semantic type that nouns denote by default (a hypothesis he terms the ‘Nominal Mapping Parameter’). It is the default denotation of nouns that is responsible for the different distributions of bare nominal arguments.
- In English, but not in Italian, members of the category NP can be kind-denoting (and so of semantic type *e*), i.e. members of the category NP are permitted to map to argument-denoting expressions.
- In a system that models the category-type association in terms of the feature **[± arg(umental)]**, English is [+arg] and Italian is [-arg].

The NP vs. DP Debate

Determined and determinerless nominals in cross-linguistic perspective

- In [-arg] Italian, members of the category NP are predicate-denoting (of semantic type $\langle e, t \rangle$) and so need to combine with a determiner to function as an argument.
- In a system that models the category-type association in terms of the feature **[$\pm$ pred(icative)]**, Italian is [+pred].
- In English, too, members of the category NP can be predicates.

- (9)
- a. The vaquita is endangered.
 - b. Adam and Zoe are doctors.

- In Chierchia's typology of NP denotations defined by [$\pm$ arg] and [$\pm$ pred], **Italian is [-arg, +pred]** and **English [+arg, +pred]**.

The NP vs. DP Debate

Determined and determinerless nominals in cross-linguistic perspective

- Excluding [-arg -pred] (which would leave nouns without any denotation), Chierchia's (1998) Nominal Mapping Parameter allows three types of languages:
 - [+arg, + pred] e.g. English
 - [-arg, + pred] e.g. Italian
 - [+arg, - pred]

The NP vs. DP Debate

Determined and determinerless nominals in cross-linguistic perspective

- In an [+arg, - pred] language, nouns can be arguments and cannot be predicates, which entails that every noun is kind-denoting. Such a language should have the following characteristics:
 - Free occurrence of bare (determinerless) arguments
 - Absence of plural morphology
 - Impossibility of combining numerals directly with nouns
 - Generalized system of classifiers
(Chierchia 1997: 78)

‘As is well known, languages with exactly these characteristics abound: Chinese, Japanese, Thai, etc.’ (Chierchia 1997: 78).

The NP vs. DP Debate

Determined and determinerless nominals in cross-linguistic perspective

Chierchia’s Nominal Mapping Parameter

	[+arg], [-pred]	[-arg], [+pred]	[+arg], [+pred]	[-arg], [-pred]
Example languages	Japanese and Chinese	French	Germanic and Slavic	
Bare nouns in argument position	Yes	No	Yes	
Morphological singular /plural distinction	No	Yes	Yes	
Generalized classifiers?	Yes	No	No	

(Punske 2014: 108)

The NP vs. DP Debate

Determined and determinerless nominals in cross-linguistic perspective

- The crucial distinction between [+ pred] and [-pred] languages lies in whether nouns require the functional category D to turn predicate-denoting nouns into nominal arguments.
- Under the Nominal Mapping Parameter hypothesis, we may then distinguish two distinct types of overtly determinerless languages:

‘Languages such as Japanese and Chinese appear to lack the semantic properties associated with D and thus potentially the functional structure. Slavic behaves in a manner similar to languages with overt Ds, so could arguably have a DP projection despite the lack of overt Ds’ (Punske 2014: 108).

The NP vs. DP Debate

An NP/DP parameter?

- What makes Slavic [+arg, + pred]?
- Czech bare plurals can be kind-denoting.
(10) Dinosauři vyhynuli.
dinosaurs became-extinct
'Dinosaurs became extinct.'
- ... as can Czech bare mass nouns.
(11) Zlato je vzácné.
gold is rare
'Gold is rare.'
- Czech also allows bare nominal predicates.
(12) Pavel je lékař.
Pavel is doctor
'Pavel is a doctor.'

The NP vs. DP Debate

An NP/DP parameter?

- Do Slavic [+arg, +pred] languages simply lack phonologically non-null exponents of D?

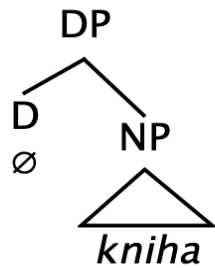
(13) a. **Kniha** je na stole.

book is on table

‘The book is on the table.’

[Czech]

b.



The NP vs. DP Debate

An NP/DP parameter?

- In a series of publications, Bošković (2004, 2005, 2008, 2009, 2012, among others) argues that there is a ‘fundamental difference in the traditional noun phrase (TNP) of languages with and languages without articles that cannot be reduced to phonology (overt vs. null articles)’ (Bošković & Gajewski 2011: 121) and that a wide range of grammatical phenomena that correlate with the presence or absence of overt articles can be explained if articleless languages lack DP altogether. In other words, Bošković proposes that the presence of a DP-layer in nominals is parameterised.

The NP vs. DP Debate

An NP/DP parameter?

Summary of syntactic properties associated with the presence or absence of overt determiners cross-linguistically (based on Bošković (2009: 199)).

	Overt D	No Overt D
Allow LBE	No	Yes
Allow scrambling	No	Yes
Can be polysynthetic	No	Yes
Allow negative raising	Yes	No
Superiority effects with multiple <i>wh</i> -fronting	Yes	No
Allow adjunct extraction from “traditional noun phrase”	No	Yes
Allow transitive nominals with two non-lexical genitives	Yes	No
Allow the majority superlative reading	Yes	No
Island sensitivity in head internal relatives	No	Yes

(Punske 2014: 115)

The NP vs. DP Debate

An NP/DP parameter?

- Left-branch extraction (cf. Ross 1967; Corver 1992)

- (14)
- a. *New_i he is buying [t_i scissors]
 - b. Nove_i on kupuje [t_i makaze]
new he buys scissors
'He is buying new scissors.'

[BCS]

(Bošković & Gajewski 2011: 122, ex. (2))

The NP vs. DP Debate

An NP/DP parameter?

- (Japanese-style) Scrambling

(15) a. ??That book, John wants to know whether Mary read.

b. Sono hon-o_i John-ga [Mary-ga e_i yonda ka dooka]siritagatteiru
that book_{ACC} John_{NOM} Mary_{NOM} read whether wants-to-know

[Japanese]

(16) a. Ovu knjigu_i Marko i Ivan znaju kada je Petar pročitao e_i.
this book Marko and Ivan know when is Petar read

[BCS]

b. ?*Kakvu knjigu_i Marko i Ivan znaju kada je Petar
what book Marko and Ivan know when is Petar
pročitao e_i?
read
'What book do Marko and Ivan know when Peter read?'

Bošković (2007, exs. (68), (69) & (72))

(17) *Dieses Buch_i, Hans will wissen, ob Maria e_i gelesen hat.
this book Hans wants know.INF whether Maria read has

[German]

The NP vs. DP Debate

An NP/DP parameter?

- What do left-branch extraction (LBE) and scrambling have to do with nominal structure?
- For **LBE**, Bošković proposes that since DP is a phase (a unit of cyclic transfer to the interfaces; see Chomsky 2000, 2001), all movement out of the DP must proceed through the phase edge, i.e. SpecDP) in accordance with the P(hase)I(mpenetrability)C(ondition).

(18) PIC

In the structure $[_{ZP} Z \dots [_{HP} \alpha [H YP]]]$

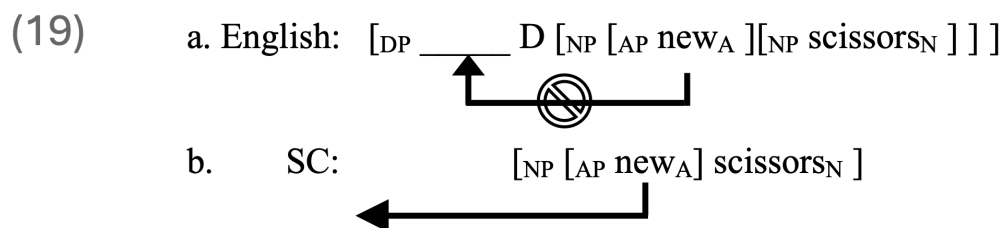
(for strong phases ZP with head Z and HP with head H),
the domain of H is not accessible to operations at ZP;
only H and its edge are accessible to such operations.

(cf. Chomsky 2001: 14)

The NP vs. DP Debate

An NP/DP parameter?

- Movement of an NP-adjoined AP to SpecDP will violate anti-locality (Grohmann 2003; Abels 2003) because it is movement that is ‘too short’, failing to cross at least one full phrasal boundary. In languages without a DP-layer, movement of AP out of the NP is not constrained by the PIC and so does not incur an anti-locality violation.



(Bošković & Gajewski 2011: 122, ex. (3))

The NP vs. DP Debate

An NP/DP parameter?

- Not all articleless languages allow LBE.

(20) *pahalı Orhan dün [~~pahalı~~ ev] aldı [Turkish]
expensive Orhan yesterday [expensive house] bought
“Orhan bought an expensive house yesterday”.
(Köylü 2021: 7, ex. (10))

The NP vs. DP Debate

An NP/DP parameter?

- Bošković's explanation of LBE in NP-languages relies on an analysis of LBE as extraction. Pereltsvaig (2008, 2013) argues against an extraction analysis of LBE, proposing instead that LBE involves copying and deletion.

(21) [**Dorogie** ~~tufli~~] on kupil [~~dorogie~~ **tufli**]. [Russian]
expensive.ACC shoes.ACC he bought expensive.ACC shoes.ACC
'It is *expensive* shoes that he bought.'

(Pereltsvaig 2013: 210, ex. (19))

The NP vs. DP Debate

An NP/DP parameter?

- Evidence against LBE as extraction is provided by LBE being able to apply to non-constituents.

- (22) a. **Protiv sovetskoj** on vystupal **vlasti**. [Russian]
against Soviet he demonstrated regime
'It is against the *Soviet* regime that he demonstrated.'
- b. Ja prosto probovala vot èti **češskie s supinatorami** pokupat' tuffi.
I simply tried here those Czech with arch-supports to-buy shoes
'I simply tried to buy those here Czech shoes with arch-supports.'

(Pereltsvaig 2013: 209, ex. (17))

The NP vs. DP Debate

An NP/DP parameter?

- Moreover, LBE appears to be sensitive to the presence of overt case marking on the fronted element.

- (23) a. {Džejms / Džejmsa} Bonda ona obožuet. [Russian]
James.NOM / James.ACC Bond.ACC she adores
'As for James Bond, she adores him.'
- b. {★Džejms / ✓Džejmsa} ona obožuet Bonda.
James.NOM / James.ACC she adores Bond.ACC
'It is *James* Bond that she adores.'

(Pereltsvaig 2013: 210, ex. (20))

The NP vs. DP Debate

An NP/DP parameter?

- Syed & Simpson (2017) criticise Bošković's explanation of LBE in NP-languages on the grounds that it requires that

‘... DP immediately dominates NP and there are no intervening functional projections. If an functional projection is projected above NP and below DP, it should be possible for an ADJP adjoined to NP to move to SpecDP and therefore potentially extract out of DP, as antilocality will not be violated in such a structure. Much argumentation has, however, been given elsewhere for the assumption that at least a NUMP / #P occurs between DP and NP (see Borer 2005 and references within this work). If this is correct, then extraction of ADJPs from DP constituents in English and other similar languages cannot be banned by anti-locality and must be attributed to some other property, which may well not involve a difference in the size of nominal constituents in a language’ (Syed & Simpson 2017: 9-10).

The NP vs. DP Debate

An NP/DP parameter?

- Hebrew, which possesses definite articles, uncontroversially qualifies as a DP-language in Bošković's typology.

(24) Ha-yeladim okhlim glidah. [Hebrew]
the-children eat.M.PL ice cream
'The children are eating ice cream.'

- As previously discussed, Ritter (1991) argues for all Hebrew noun phrases to contain a NumP. If so, it is predicted under Bošković's account that Hebrew allows LBE since no violation of anti-locality will arise when an AP adjoined to NP raises to SpecDP. This prediction is not borne out.

(25) a. Hu kone misparayim khadashim.
he buys scissors new.M.PL
'He is buying new scissors'

b. *Khadashim hu kone misparayim.
new.M.PL. he buys scissors

The NP vs. DP Debate

Evidence for DP-structure in articleless languages

- In various languages without articles, word order alternations within nominals produce different interpretations, which may suggest that even in articleless languages, DP can be identified as a locus of referentiality.

The NP vs. DP Debate

Evidence for DP-structure in articleless languages

- In **Bangla**, the canonical sequencing of *numeral* > *classifier* > *adjective* > *noun* produces an indefinite interpretation of nominal phrases. The inverted order of *adjective* > *noun* > *numeral* > *classifier* gives rise to a definite interpretation.

- (27) a. du ʈo lal boi (canonical order: NUM > CL > ADJ > NOUN) [Bangla]
 two CL red book
 ‘two red books’
- b. lal boi du ʈo (inverted order: ADJ > NOUN > NUM > CL)
 red book two CL
 ‘the two red books’

(Syed & Simpson 2017: 2, exs (1) & (2))

The NP vs. DP Debate

Evidence for DP-structure in articleless languages

- The Russian and Bangla data receive an explanation if leftward movement of the demonstrative (in Russian) or the phrasal complement of the classifier (in Bangla) targets the DP-domain, which is associated with definiteness.

The NP vs. DP Debate

Evidence for DP-structure in articleless languages

- However, this would mean that Russian and Bangla, both of which are articleless languages, must project a DP-layer and so are DP-languages in Bošković's classification.
- If Russian is a DP-language, it falsifies Bošković's claim that only NP-languages license LBE.

The NP vs. DP Debate

Evidence for DP-structure in articleless languages

- At first blush, Bangla seems to show the syntactic behaviour expected for DP-languages and disallow LBE.

(28) *Dami_i se t_i boi ta kinlo. [Bangla]
Expensive he book CL bought.
(Syed & Simpson 2017: 9, ex (27))

- But LBE can be fully acceptable in Bangla as long as it is accompanied by appropriate focus intonation or stress.

(29) Lal_i oi [t_i boi [a] chai.
Red DEM book CL want.
'I want that RED book'
(Syed & Simpson 2017: 9, ex (28))

The NP vs. DP Debate

Evidence for DP-structure in articleless languages

- The variable behaviour of Bangla with respect to LBE may be interpreted as strengthening Pereltsvaig's (2008, 2013) claim that LBE in articleless languages does not in fact involve extraction. This would remove LBE as a diagnostic for distinguishing DP- and NP-languages and thereby surmount the empirical challenge of two articleless languages displaying divergent syntactic behaviour concerning LBE.

The NP vs. DP debate

Evidence for DP-structure in articleless languages

Try your hand at some data analysis

- Consider the following BCS data (from Progovac 1998).

- (30)
- I samu Mariju to nervira*
and alone-ACC Mary-ACC that-NOM irritates
'That irritates even Mary.'
 - ?*I Mariju samu to nervira*
and Mary.ACC alone-ACC that-NOM irritates
 - I nju samu to nervira.*
and her-ACC alone-ACC that-NOM irritates
'That irritates even her.'
 - ?*I-samu nju to nervira.*
and alone-ACC her-ACC that-NOM irritates

Question 1: What is the relative ordering of nominals and their modifiers?

Question 2: Assuming that the adjectival modifiers occupy a fixed position within the nominal phrase, how can you account for the relative orderings observed?

Question 3: BCS is an articleless language. Do the data provide evidence for it being a DP- or an NP-language?

The NP vs. DP debate

Evidence for DP-structure in articleless languages

- While BSC nouns must follow an intensifying adjective that modifies them, pronouns must precede the intensifying adjective.
- On the assumption that intensifying adjectives occupy a fixed position within the nominal phrase, the patterns observed suggest that pronouns occupy a structurally higher position than nouns.
- In Progovac's (1998) analysis, both pronouns and nouns are merged in N, and pronouns undergo N-to-D movement, i.e. BSC is hypothesised to project a DP.
If we understand NP to be separated from DP by any number of functional categories, additional evidence would be needed to justify the claim that pronouns target D.

The NP vs. DP debate

Summary and outlook

- Developing a *semantic* typology of languages, Chierchia (1998) argues that bare NPs can function directly as arguments in [+arg] languages, i.e. argumenthood does not universally require a DP-structure.
- Bošković (2004 and subsequent work) posits a *syntactic* parameter that distinguishes NP- from DP-languages, arguing that a cluster of diverse syntactic properties (including the availability of LBE) stems from a single parameter setting.
- In Progovac's (1998) analysis, BCS personal pronouns are merged in N and raise to D. How do the English data below broaden the perspective on the categorical status of pronouns?

- (31)
- | | |
|--------------------|--------------------|
| a. we linguists | c. us linguists |
| b. *they linguists | d. ?them linguists |

The NP vs. DP debate

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The NP vs. DP debate

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